

Business Requirements Documentundefined

This document serves as the authoritative specification for version 1. It defines the business needs, functional requirements, and success criteria necessary for successful project delivery. This document is intended for use by stakeholders, project managers, developers, and quality assurance teams.

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Executive Summary

This document outlines the business requirements for 'VeriTour', a Smart Tourist Safety Monitoring & Incident Response System. VeriTour leverages AI, Blockchain, and Geo-Fencing technologies to enhance the safety of tourists in regions like Northeast India. The system's core components include a secure blockchain-based Digital Tourist ID, a multi-functional mobile application for tourists with a panic button and safety score, an AI-powered anomaly detection engine, and a unified real-time dashboard for police and tourism departments to facilitate rapid and coordinated incident response.

Project Overview

In key tourism regions, traditional safety and tracking methods are often insufficient, especially in remote or high-risk areas. VeriTour addresses this challenge by creating a comprehensive digital ecosystem. The system will issue secure, tamper-proof digital IDs to tourists upon entry, valid for their visit's duration. A companion mobile application will provide tourists with geo-fencing alerts for restricted zones, a dynamic safety score, and a one-touch panic button. An AI engine will monitor for anomalies like sudden location drop-offs or route deviations. A web-based dashboard will provide authorities with real-time situational awareness, heat maps of high-risk zones, and tools for incident management, including automated E-FIR generation. The project aims to provide a robust, technology-driven solution that ensures real-time monitoring, rapid response, and secure identity verification while respecting user privacy.

Business Objectives

- To enhance the safety and security of tourists in key regions.
- To provide real-time monitoring of tourist locations and potential risks.
- To significantly reduce incident response times through automated alerts and a unified command dashboard.
- To provide a secure, tamper-proof, and verifiable digital identity system for tourists.
- To improve coordination and operational efficiency between tourism and police departments.
- To build trust and promote tourism by establishing a reputation for safety.

Scope and Boundaries

```
{  
  "in_scope": [  
    "Development of a blockchain-based Digital Tourist ID generation and verification system.",  
    "Development of a cross-platform (iOS & Android) mobile application for tourists.",  
    "Implementation of features within the mobile app: Tourist Safety Score, Geo-fencing alerts, Panic Button, and  
Opt-in real-time tracking.",  
    "Development of an AI/ML engine for anomaly detection (location drop-offs, prolonged inactivity, route  
deviation).",  
    "Development of a web-based dashboard for Tourism Department and Police with real-time visualizations, heat  
maps, and incident management tools.",  
    "Implementation of an automated E-FIR generation workflow for missing person cases.",  
    "Multilingual support for the mobile app and web portals (10+ Indian languages and English).",  
    "Optional integration of IoT wearables (smart bands/tags) for enhanced safety in high-risk areas."  
  ],  
  "out_of_scope": [  
    "Integration with commercial hotel booking, travel reservation, or payment systems.",  
    "Development of general tourist guide functionalities (e.g., points of interest, restaurant reviews) beyond safety  
information.",  
    "Management of visa or immigration processing; the system will only consume verified data.",  
    "Billing and financial management systems for the service itself."  
  ]  
}
```

Stakeholder Analysis

- name: Tourists
 - role: Primary End-Users
 - responsibility: Use the mobile application for safety, activate their Digital ID, and utilize features like the Panic Button and location sharing.
-

- name: Police Units
 - role: Primary Responders
 - responsibility: Monitor the dashboard for alerts, respond to incidents, manage E-FIRs, and utilize the system for investigations.
-

- name: Tourism Department Personnel
 - role: System Operators & Policy Makers
 - responsibility: Monitor overall tourist safety, manage high-risk zones, analyze safety data and trends, and coordinate with police.
-

- name: Entry Point Staff (Immigration, Hotel, etc.)
 - role: Onboarding Facilitators
 - responsibility: Use the Entry Point Portal to securely issue and verify Digital Tourist IDs.
-

- name: Government Ministries (Home Affairs, Tourism, External Affairs)
 - role: Project Sponsors & Consortium Members
 - responsibility: Provide governance, legal frameworks, access to necessary data/APIs, and operate blockchain nodes.
-

- name: Emergency Contacts / Family Members
 - role: Secondary Users
 - responsibility: Receive alerts if designated by a tourist and view shared location data if the tourist opts-in.
-

- name: System Administrators
- role: Technical Management

- responsibility: Manage system infrastructure, user access, security, and maintenance of all technical components.
-

Functional Requirements

- The system shall issue a secure, blockchain-based Digital ID to tourists at designated entry points.

- The Digital ID shall store KYC information, trip itinerary, and emergency contacts, and be valid only for the trip duration.

- The mobile app shall calculate and display a dynamic 'Tourist Safety Score' based on location, itinerary, and area sensitivity.

- The mobile app shall provide real-time geo-fencing alerts when a tourist enters a pre-defined high-risk or restricted zone.

- The mobile app shall feature a prominent Panic Button that shares live location with the nearest police unit and emergency contacts.

- The mobile app shall offer an explicit opt-in feature for real-time location sharing with designated contacts.

- The AI engine shall detect anomalies such as sudden location drop-offs, prolonged inactivity, and significant deviation from planned routes.

- The authority dashboard shall provide real-time map-based visualizations of tourist clusters and heat maps of high-risk zones.

- The dashboard shall provide authorized personnel with secure access to tourist Digital ID records, alert history, and last known locations.

- The dashboard shall support the automated generation of a pre-filled Electronic First Information Report (E-FIR) for missing person cases.

- The system shall support optional IoT smart bands that transmit continuous location and health signals and have a manual SOS button.

-
- The mobile app and web portals shall support at least 10 Indian languages in addition to English.
-

Non-Functional Requirements

- All data in transit and at rest shall be protected with end-to-end encryption.

- The system must comply with India's Digital Personal Data Protection Act (DPDPA) and other relevant data privacy laws.

- The system must maintain high availability (99.9% uptime) for all critical monitoring and alerting functions.

- The system architecture must be scalable to handle a large and growing number of tourists.

- The target SLA for alert dispatch to the police dashboard is less than 30 seconds from trigger.

- The target SLA for police acknowledgment of a critical alert is less than 60 seconds.

- The mobile application must be intuitive and easy to use, especially the Panic Button feature.

- The system must ensure data integrity and auditability through the use of a permissioned blockchain for ID issuance and incident logging.

- The system shall be resilient to network failures, with offline data buffering and fallback communication channels (e.g., SMS).

- The authority dashboard must render map data and real-time updates with minimal latency.

User Requirements

- user_role: Tourist
 - action: press a single, prominent SOS button in the app
 - goal: to quickly alert authorities and my emergency contacts in an emergency.
 - acceptance_criteria: A 5-second cancellation window is provided after pressing the button.,A confirmation message is displayed after the alert is successfully sent.,The app provides status updates on the response (e.g., 'Police Acknowledged').
-

- user_role: Police Officer
 - action: view a tourist's complete profile, live location, and incident context on a single screen
 - goal: to have all necessary information for a rapid and effective response.
 - acceptance_criteria: The view includes a photo, ID details, emergency contacts, blood type, and last known location.,The location updates in real-time on an interactive map.,An incident timeline logs all events related to the alert.
-

- user_role: Tourism Official
 - action: view a real-time heat map of tourist density and high-risk zones
 - goal: to proactively manage tourist safety, issue advisories, and allocate resources.
 - acceptance_criteria: The map displays color-coded zones based on dynamic risk scores.,The map shows anonymized tourist clusters to gauge density.,The official can filter map layers for different types of information.
-

- user_role: Family Member
 - action: view my relative's live location on a secure web portal
 - goal: to have peace of mind during their travels, provided they have opted-in to share.
 - acceptance_criteria: Access is granted only through a secure, temporary link sent via SMS/notification.,The portal is view-only and shows only the tourist's location and name.,Access automatically expires when the sharing duration ends.
-

User Stories and Use Cases

- Tourist Digital ID Generation & Onboarding: A tourist arrives at an entry point, provides their passport/Aadhaar, and receives a secure Digital ID linked to their mobile app.

- Panic Button Activation & Emergency Response: A tourist in distress activates the panic button, triggering an automated, multi-channel alert to the nearest police unit and their emergency contacts.

- Opt-in Real-time Tracking & Management: A tourist chooses to share their live location with specific family members for a defined period for added peace of mind.

- AI Anomaly Detection and Investigation: The system flags a tourist for prolonged inactivity in a remote area, prompting an operator to investigate.

- Zone Management by Tourism Officials: An authorized official uses the dashboard to draw a new temporary high-risk geo-fence due to a landslide warning.

- E-FIR Generation for Missing Person: Following a confirmed missing person report, a police officer uses the dashboard to automatically generate a pre-filled E-FIR.

Business Rules

- A Digital Tourist ID is valid only for the duration of the tourist's declared trip itinerary and expires automatically.

- A Panic Button activation includes a 5-second cancellation window to mitigate false alarms.

- An alert not acknowledged by the primary nearest police unit within 2 minutes must be automatically escalated to a wider radius and senior officials.

- Law enforcement access to a tourist's location data (outside of an active panic alert) requires the tourist's explicit opt-in or a verified E-FIR number as a legal trigger.

- Location data from family sharing is purged immediately after the sharing session ends or the trip concludes.

- Incident data related to an emergency is retained in an encrypted archive for 7 years for legal and audit purposes.

Assumptions and Dependencies

```
{  
  "assumptions": [  
    "The majority of tourists will possess and use a smartphone with GPS and data capabilities.",  
    "Tourists will consent to the terms of service, including location monitoring for safety purposes.",  
    "Government agencies (Police, Tourism, Immigration) will cooperate by providing necessary data sharing agreements and API access.",  
    "Entry point staff can be effectively trained to operate the ID issuance portal."  
  ],  
  "dependencies": [  
    "Availability and reliability of government APIs for KYC verification (e.g., DigiLocker, Passport Seva, IVFRT).",  
    "Availability of a secure API for E-FIR submission and verification with state police systems.",  
    "Access to real-time environmental and weather data from the India Meteorological Department (IMD) API.",  
    "Reliable and widespread cellular and internet connectivity in tourist areas for the system to function effectively.",  
    "Partnerships with telecom providers for reliable SMS and voice call alert delivery.",  
    "Formal establishment of a blockchain consortium with participation from key government ministries."  
  ]  
}
```

Success Criteria

- Achieve an average incident response time (from alert trigger to police acknowledgment) of under 90 seconds.

- Achieve a 98% or higher SLA compliance for alert dispatch (<30s) and acknowledgment (<60s).

- Attain a tourist adoption rate of over 75% in pilot regions within the first year of rollout.

- Demonstrate a measurable reduction in the number of unresolved tourist safety incidents in deployed areas.

- Receive positive feedback scores (above 4/5) from police and tourism department users on the dashboard's effectiveness and usability.

- Successful completion of a pilot program in a designated region with positive outcomes and stakeholder approval.

Risk Assessment

- risk: Low adoption rate among tourists due to privacy concerns or perceived complexity.
 - impact: High
 - mitigation: Implement a 'Privacy by Design' approach with clear, transparent communication. Conduct awareness campaigns highlighting safety benefits. Ensure a simple, user-friendly onboarding process.
-

- risk: A data privacy breach leading to the exposure of sensitive tourist information (PII).
 - impact: Critical
 - mitigation: Enforce end-to-end encryption, strict role-based access control, secure off-chain storage for PII, and regular security audits and penetration testing. Ensure full DPDPA compliance.
-

- risk: Unavailability or poor performance of critical government APIs (KYC, FIR).
 - impact: High
 - mitigation: Develop robust fallback mechanisms, including offline issuance modes with deferred validation and manual override procedures with audit trails. Establish strong service level agreements with government partners.
-

- risk: High rate of false alarms from the panic button draining emergency resources.
 - impact: Medium
 - mitigation: Implement a 5-second cancellation window. Design the panic button UI to prevent accidental presses. Flag accounts with frequent false alarms for a mandatory tutorial review.
-

- risk: Inaccurate AI anomaly detection leading to false positives or missed incidents.
 - impact: High
 - mitigation: Train models on extensive and diverse datasets. Employ a 'human-in-the-loop' system for reviewing high-confidence alerts before dispatch. Continuously evaluate and retrain models based on real-world outcomes.
-

Timeline and Milestones

- milestone: Phase 1: Foundation & MVP Development
 - date: Months 1-9
 - deliverable: A functional MVP including the core mobile app (ID activation, Panic Button), the Entry Point Portal for ID issuance, and a basic Authority Dashboard for monitoring panic alerts.
-

- milestone: Phase 2: Enhanced Safety & Operations
 - date: Months 10-16
 - deliverable: Integration of AI-based Anomaly Detection, dynamic heat maps, full incident management features including E-FIR generation, and advanced opt-in tracking for tourists.
-

- milestone: Phase 3: IoT Integration & Optimization
 - date: Months 17-21
 - deliverable: Full support for optional IoT smart bands/tags, system-wide performance optimization, and development of advanced analytics and reporting modules.
-

- milestone: Phase 4: Pilot Deployment & Rollout
 - date: Months 22-24+
 - deliverable: Successful completion of a controlled pilot program in a designated region, comprehensive user training, and initiation of a phased, region-by-region rollout of the full system.
-

Resource Requirements

- resource_type: Project Management

- quantity: 1-2

- skills: Agile methodologies, Stakeholder Management, Risk Management

- resource_type: UI/UX Design

- quantity: 2-3

- skills: Mobile & Web App Design, Prototyping (Figma/Sketch), Data Visualization, Usability Testing

- resource_type: Mobile App Development

- quantity: 4-6

- skills: Flutter or React Native, iOS (Swift), Android (Kotlin), Secure Coding, GPS/Geolocation APIs

- resource_type: Web Frontend Development

- quantity: 3-4

- skills: React.js/Next.js, Data Visualization Libraries (D3.js), Mapbox GL JS, State Management

- resource_type: Backend Development

- quantity: 6-8

- skills: Microservices, Node.js/Go/Python, API Design (REST/GraphQL), Docker, Kubernetes, Kafka/Kinesis

- resource_type: Blockchain Development

- quantity: 2-3

- skills: Hyperledger Fabric, Chaincode (Go/Node.js), Consortium Network Setup

- resource_type: AI/ML Engineering

- quantity: 3-4

- skills: Python, TensorFlow/PyTorch, Anomaly Detection, Geospatial Data Analysis, MLOps

- resource_type: DevOps/Cloud Engineering

- quantity: 2-3

- skills: AWS/Azure/GCP, CI/CD Pipelines, Infrastructure as Code (Terraform), Security, Monitoring

- resource_type: Quality Assurance

- quantity: 3-5

- skills: Manual & Automated Testing, Performance Testing, Security Testing

Approval and Sign-off

- stakeholder: Representative, Ministry of Tourism
 - role: Project Sponsor / Consortium Member
 - approval_required: Approval of business objectives, functional requirements, and pilot program scope.
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- stakeholder: Representative, Ministry of Home Affairs (Police)
 - role: Primary User / Consortium Member
 - approval_required: Approval of incident response workflows, dashboard functionality, and E-FIR integration.
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- stakeholder: Representative, Ministry of External Affairs (Immigration)
 - role: Consortium Member
 - approval_required: Approval of Digital ID generation process and integration with immigration/visa data systems.
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- stakeholder: Project Sponsor / Lead
 - role: Overall Project Owner
 - approval_required: Final sign-off on the complete Business Requirements Document.
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